PAIRS WITH LOCALLY NILPOTENT IDEAL ARE HENSELIAN AND ETALE ALGEBRAS REDUCE BIJECTIVELY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Editorial note. Benign-screen note: the proof contains the phrase 'it is clear that' / 'is obvious' once, used as a legitimate step inside a complete argument; the author proves the statement, so this is not a hint-only proof.

Allowed standard prerequisites (editorial).

- Lifting of etale algebras, and the strong lifting property for smooth algebras. Algebra, Lemma \ref{algebra-lemma-lift-etale} and Algebra, Lemma \ref{algebra-lemma-smooth-strong-lift}, cited at proof lines 2424 and 2428-2429. Give essential surjectivity and fullness.
- An etale algebra is unramified, and the kernel of the multiplication map is generated by an idempotent. Algebra, Lemma \ref{algebra-lemma-diagonal-unramified} and Algebra, Lemma \ref{algebra-lemma-unramified}, cited at proof lines 2433-2435. Provide the idempotent used for faithfulness.
- Idempotents lift modulo a locally nilpotent ideal, and the ideal is locally nilpotent after base change. Algebra, Lemma \ref{algebra-lemma-lift-idempotents} and Algebra, Lemma \ref{algebra-lemma-locally-nilpotent}, cited at proof lines 2437-2438. Give faithfulness and support the radical argument.
- Lifting of a factorisation of a monic polynomial into monic factors after an etale base change inducing an isomorphism modulo the ideal. Lemma \ref{lemma-lift-factorization-monic}, cited at proof lines 2445-2447. Provides the factorisation used in the henselian verification.

Source reference. Original lemma 0ALI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13371–13371. The lemma's own canonical tag,

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quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 04D1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5711–5711. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07K4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9644–9644. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 02FL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3196–3196. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00UV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1125–1125. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0544: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6627–6627. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00J9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 707–707. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0ALH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13370–13370. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let (A, I) be a pair with I locally nilpotent. Then the functor $B \mapsto B/IB$ induces an equivalence between the category of étale algebras over A and the category of étale algebras over A/I. Moreover, the pair is henselian.*

Proof. Essential surjectivity holds by Algebra, Lemma 04D1. If B, B' are étale over A and $B/IB \to B'/IB'$ is a morphism of A/I-algebras, then we can lift this by Algebra, Lemma 07K4. Finally, suppose that $f, g : B \to B'$ are two A-algebra maps with $f \bmod I = g \bmod I$. Choose an idempotent $e \in B \otimes_A B$ generating the kernel of the multiplication map $B \otimes_A B \to B$, see Algebra, Lemmas 02FL and 00UV (to see that étale is unramified). Then $(f \otimes g)(e) \in IB'$. Since IB' is locally nilpotent (Algebra, Lemma 0544) this implies $(f \otimes g)(e) = 0$ by Algebra, Lemma 00J9. Thus $f = g$.

It is clear that I is contained in the Jacobson radical of A. Let $f \in A[T]$ be a monic polynomial and let $\overline{f} = g_0 h_0$ be a factorization of $\overline{f} = f \bmod I$ with $g_0, h_0 \in A/I[T]$ monic generating the unit ideal in $A/I[T]$. By Lemma 0ALH there exists an étale ring map $A \to A'$ which induces an isomorphism $A/I \to A'/IA'$ such that the factorization lifts to a factorization into monic polynomials over A'. By the above we have $A = A'$ and the factorization is over A. □